Dirichlet Characters and Gauss Sums

Given a positive integer m, a map $\chi : \mathbb{Z} \to \mathbb{C}$ which has the following properties is called a *Dirichlet character with modulus* m:

1. $\chi(a \cdot b) = \chi(a) \cdot \chi(b)$.
2. $\chi(a + m) = \chi(a)$.
3. $\chi(a) \neq 0$ if and only if $\gcd(a, m) = 1$.

Given ζ, a primitive m-th root of unity, we define the *Gauss Sum associated with the Dirichlet character* χ *of modulus* m as

$$G(\chi) = \sum_{a=1}^{m} \chi(a)\zeta^a$$

As it stands, this appears to depend on the choice of ζ, but in some sense, since all choices may appear to be equivalent under a Galois automorphism, it does not!

For example, if we take the *natural* primitive m-th root $\exp(2\pi\sqrt{-1}/m)$ in $\mathbb{C}$ in place of ζ, then

$$G(\chi) = \sum_{a=1}^{m} \chi(a) \exp\left(2\pi\sqrt{-1}\frac{a}{m}\right)$$

If $\gcd(k, m) = 1$, then we can also take

$$G(\chi) = \sum_{a=1}^{m} \chi(a) \exp\left(2\pi\sqrt{-1}\frac{ak}{m}\right)$$

We use the expression

$$G(\chi, k) = \sum_{a=1}^{m} \chi(a) \exp\left(2\pi\sqrt{-1}\frac{ak}{m}\right)$$

which makes it clear which *complex* number we are taking the Gauss sum to be. Moreover, this expression has a meaning even when k is *not* co-prime to m.

We will especially be interested in Gauss sums associated with a Dirichlet character of prime modulus.

Basic results about Dirichlet Characters

In modern language, we can say that a Dirichlet character is uniquely determined by a group homomorphism

$$\chi_* : \left(\frac{\mathbb{Z}}{\langle m \rangle}\right)^* \to \mathbb{C}^*$$

by defining

$$\chi(a) = \begin{cases} \chi_*(a \bmod m) & \gcd(a, m) = 1 \\ 0 & \gcd(a, m) \neq 1 \end{cases}$$

Since χ_* is a group homomorphism from a finite group of order $\phi(m)$, its image is contained in the group of $\phi(m)$-th roots of unity. If $\gcd(a,m) = 1$ and a has (multiplicative) order k modulo m, then $\chi(a)$ is a k-th root of unity.

We see that Dirichlet characters with modulus m form a group which can be identified as above with the group

$$\mathrm{Hom}((\mathbb{Z}/\langle m\rangle)^*, \mathbb{C}^*)$$

However, it is useful to remember the extension $\chi(a) = 0$ when $\gcd(a,m) \neq 1$.

The "trivial" homomorphism is associated with the *principal Dirichlet character with modulus m*:

$$\chi_{0,m}(a) = \begin{cases} 1 & \gcd(a,m) = 1 \\ 0 & \gcd(a,m) \neq 1 \end{cases}$$

This is the "characteristic function" of the set of integers co-prime to m. Note that, as a function, it depends only on the primes dividing m.

The product (in the above group structure) of Dirichlet characters χ_1 and χ_2 of modulus m is defined by:

$$(\chi_1 \cdot \chi_2)(a) = \chi_1(a) \cdot \chi_2(a)$$

More generally, if χ_1, χ_2 are Dirichlet characters with modulus m_1, m_2 respectively, then $(\chi_1 \cdot \chi_2)$ as defined above gives a Dirichlet character with modulus m, where m is the least common multiple of m_1 and m_2.

Since non-zero values of a Dirichlet character are complex numbers of absolute value 1, we see that the inverse (in the group law) of a Dirichlet character χ is defined by:

$$\overline{\chi}(a) = \overline{\chi(a)}$$

so that $\chi \cdot \overline{\chi} = \chi_{0,m}$ is the identity element in the group structure on Dirichlet characters of modulus m.

Decomposition

If $m = m_1 \cdot m_2$, where $\gcd(m_1, m_2) = 1$, then (by the Chinese Remainder Theorem) we have:

$$\left(\frac{\mathbb{Z}}{\langle m\rangle}\right)^* = \left(\frac{\mathbb{Z}}{\langle m_1\rangle}\right)^* \times \left(\frac{\mathbb{Z}}{\langle m_2\rangle}\right)^*$$

Note that $(\mathbb{Z}/\langle m_1\rangle)^*$ is being identified with the subgroup of $(\mathbb{Z}/\langle m\rangle)^*$ consisting of elements that are congruent to 1 modulo m_2. A similar statement holds with with m_1 and m_2 interchanged.

We can make this isomorphism explicit using r_1, r_2 such that $r_1 m_1 + r_2 m_2 = 1$. We put $e_1 = r_2 m_2$ and $e_2 = r_1 m_1$. We have the identities:

$$\begin{aligned} e_1 \bmod m_1 &= 1 & \qquad e_1 \bmod m_2 &= 0 \\ e_2 \bmod m_1 &= 0 & \qquad e_2 \bmod m_2 &= 1 \end{aligned}$$

It follows that for every pair of integers a and b, the integer ae_1+be_2 is congruent to a modulo m_1 and is congruent to b modulo m_2. Moreover,

$$(ae_1 + be_2) \bmod m = (ae_1 + e_2)(e_1 + be_2) \bmod m$$

and

$$\begin{aligned}(ae_1 + be_2) \bmod m &= ((a + m_1)e_1 + be_2) \bmod m\\ &= (ae_1 + (b + m_2)e_2) \bmod m\end{aligned}$$

and

$$\begin{aligned}\gcd(a, m_1) &= \gcd(ae_1 + e_2, m_1)\\ \gcd(b, m_2) &= \gcd(e_1 + be_2, m_2)\end{aligned}$$

Given a Dirichlet character χ with modulus m we define $\chi_1(a) = \chi(ae_1 + e_2)$. Similarly, we define $\chi_2(a) = \chi(e_1 + ae_2)$.

Using the above calculations, we check that χ_i is a Dirichlet character with modulus m_i. Moreover, we see that $\chi(a) = \chi_1(a)\chi_2(a)$ for every integer a.

Conversely, given Dirichlet characters χ_i with modulus m_i with $\gcd(m_1, m_2) = 1$, we define

$$\chi(a) = \chi_1(a)\chi_2(a)$$

As seen above, this is a Dirichlet character of modulus $m = m_1 \cdot m_2$ which is the least common multiple of m_1 and m_2. More directly, we note that $\gcd(a, m) = \gcd(a, m_1)\gcd(a, m_2)$. Hence $\gcd(a, m) = 1$ if and only if $\gcd(a, m_1) = 1$ and $\gcd(a, m_2) = 1$. This shows that $\gcd(a, m) \neq 1$ if and only if $\chi(a) = 0$.

If $m = p_1^{k_1} \cdots p_r^{k_r}$ is the factorisation of m into prime powers, one can use induction to write every Dirichlet character with modulus m as a product of Dirichlet characters χ_i with modulus $p_i^{k_i}$ for $i = 1, \ldots, k$.

Thus, the structure of Dirichlet characters can be reduced to the structure of Dirichlet charcters with modulus a prime power.

Conductors and primitive Dirichlet characters

Given positive integers m and d such that d divides m, we have a natural group homomorphism

$$\left(\frac{\mathbb{Z}}{\langle m\rangle}\right)^* \to \left(\frac{\mathbb{Z}}{\langle d\rangle}\right)^*$$

Given a group homomorphism

$$\rho_* : \left(\frac{\mathbb{Z}}{\langle d\rangle}\right)^* \to \mathbb{C}^*$$

we get a group homomorphism

$$\chi_* : \left(\frac{\mathbb{Z}}{\langle m\rangle}\right)^* \to \mathbb{C}^*$$

by composition. Thus, we have Dirichlet characters χ and ρ with modulus m and d. What is the relation between these?

Clearly, if $\gcd(a, m) = 1$, then $\chi(a) = \rho(a)$. However, if $\gcd(a, d) = 1$ and $\gcd(a, m) \neq 1$ (this is the case if a has a prime factor p which divides m/d but not d), then $\chi(a) = 0$ whereas $\rho(a) \neq 0$.

Recall that $\chi_{0,m}$ denotes the principal Dirichlet character with modulus m; this is the characteristic function of the set of integers co-prime to m.

We easily check the identity $\chi(a) = \rho(a)\chi_{0,m}(a)$ for all integers a.

Conversely, given such a relation $\chi = \rho\chi_{0,m}$ between a Dirichlet character χ with modulus m and a Dirichlet character ρ with modulus d, we see easily that the group homomorphism χ_* factors via ρ_* as above.

As noted above, $\chi_{0,m}$ depends only on the prime factors of m. Thus, if p is a prime factor m, then a Dirichlet character of modulus m is also a Dirichlet character of modulus $p \cdot m$. However, the converse need not be true. The conductor of a Dirichlet character, as defined below, removes "irrelevant" prime factors from m.

When $m = p^a$ for a prime p, we see that $d = p^b$ for some $b \leq a$. It is clear that there is a *smallest* b such that χ can be expressed in terms of a ρ as above. (For example, $b = 0$ if and only if χ is the principal Dirichlet character χ_{0,p^a}.)

Using the prime decomposition $m = p_1^{a_1} \cdots p_r^{a_r}$ we have a decomposition of a Dirichlet character χ of modulus m in terms of Dirichlet characters χ_i of modulus $p_i^{a_i}$ for $i = 1, \ldots, r$. Thus, we see that there is a *smallest* positive integer divisor $c = p_1^{b_1} \cdots p_r^{b_r}$ of m such that there is a Dirichlet character ρ of modulus c so that $\chi(a) = \rho(a)\chi_{0,m}(a)$ for all integers a. This c is called the *conductor* of the Dirichlet character χ with modulus m.

If there was a factor d of c and a Dirichlet character τ with modulus d such that $\rho(a) = \tau(a)\chi_{0,c}(a)$ for all integers a, then $\chi(a) = \tau(a)\chi_{0,c}(a)\chi_{0,m}(a)$ for all integers a. Note that $\chi_{0,c}\chi_{0,m} = \chi_{0,m}$ since c divides m. This means that $\chi(a) = \tau(a)\chi_{0,m}(a)$ for all integers a. Since c is the smallest factor of m for which such an identity holds, we see that $d = c$.

In other words, we have shown that ρ is a Dirichlet character with modulus c such that it cannot be further expressed in terms of a Dirichlet character for a proper factor of c. Such a character is called a *primitive* Dirichlet character with modulus c.

Warning: Note that this *does not* necessarily mean that the group homomorphism $(\mathbb{Z}/\langle c\rangle)^* \to \mathbb{C}^*$ is injective! This is why we distinguish between characters of finite abelian groups and Dirichlet characters.

In summary, we have shown that any Dirichlet character χ with modulus m can be written in the form $\chi = \rho\chi_{0,m}$ where ρ is a primitive Dirichlet character with modulus c which is the conductor of χ.

Dirichlet characters with modulus odd prime power

Given an odd prime p.

As seen earlier, the unit group $U(\mathbb{Z}_p)$ of $\mathbb{Z}_p$ is isomorphic to $\mathbb{F}_p^* \times \mathbb{Z}_p$ via the map $\mathbb{F}_p^* \times \mathbb{Z}_p \to U(\mathbb{Z}_p)$ given by $(a, x) \mapsto \omega(a)\exp(px)$.

Since the ring homomorphism $\mathbb{Z}_p \to \mathbb{Z}/\langle p^k \rangle$ is surjective and $\mathbb{Z}_p$ is a local ring, we see that the group $(\mathbb{Z}/\langle p^k \rangle)^*$ is isomorphic to $\mathbb{F}_p^* \times \mathbb{Z}_p/\langle p^{k-1} \rangle$.

The group $(\mathbb{Z}/\langle p^k \rangle)^*$ is a cyclic group of order $(p-1)p^{k-1}$.

More explicitly, let ζ_{p-1} be a primitive $(p-1)$-th root of unity in $\mathbb{Z}_p$. The element $g_p = \zeta_{p-1}\exp(p)$ generates the above group.

It follows that a Dirichlet character χ with modulus p^k is determined by the image $\chi(g_p)$. The latter is of the form $\omega_{p^{k-1}}^u$ where $\omega_{p^{k-1}}$ is a primitive p^{k-1}-th root of unity and u is an element of $\mathbb{Z}/\langle (p-1)p^{k-1} \rangle$.

Note that we have a decomposition

$$\frac{\mathbb{Z}}{\langle (p-1)p^{k-1} \rangle} = \frac{\mathbb{Z}}{\langle p-1 \rangle} \times \frac{\mathbb{Z}}{\langle p^{k-1} \rangle}$$

We easily check, for $k > 1$, that χ is primitive if and only if it is injective on the subgroup $\mathbb{Z}/\langle p^{k-1} \rangle$. For $k = 1$, every non-principal character of modulus p is primitive. Note that for $p \geq 5$ a non-principal character of modulus p need not be injective on $\mathbb{F}_p^*$. For example, we have a natural surjective homomorphism $\mathbb{F}_p^* \to \{\pm 1\}$ for any odd prime. This gives a primitive Dirichlet character of modulus p.

Dirichlet characters with modulus 2^k

As seen earlier, the unit group $U(\mathbb{Z}_2)$ of $\mathbb{Z}_2$ is isomorphic to $\{\pm 1\} \times \mathbb{Z}_2$ via the map $\{\pm 1\} \times \mathbb{Z}_2 \to U(\mathbb{Z}_2)$ given by $(s, x) \mapsto s\exp(4x)$.

Note that $(\mathbb{Z}/\langle 2 \rangle)^*$ is a trivial group. Thus, there is *no* primitive Dirichlet character with modulus $2 = 2^1$. For $k \geq 2$ we proceed as follows.

Since the ring homomorphism $\mathbb{Z}_2 \to \mathbb{Z}/\langle 2^k \rangle$ is surjective and $\mathbb{Z}_2$ is a local ring, we see that the group $(\mathbb{Z}/\langle 2^k \rangle)^*$ is isomorphic to $\{\pm 1\} \times \mathbb{Z}_2/\langle 2^{k-2} \rangle$.

Note that when $k > 2$ the group is *not* cyclic. It is generated by -1 and $g_2 = \exp(4)$.

A Dirichlet character χ with modulus 2^k is determined by the images $\chi(-1)$ and $\chi(g_2)$. Clearly, $\chi(-1) = \pm 1$, while $\chi(g_2)$ is of the form $\omega_{2^{k-2}}^u$, where $\omega_{2^{k-2}}$ is a primitive 2^{k-2}-th root of unity and u is an element of $\mathbb{Z}/\langle 2^{k-2} \rangle$.

In this case, for $k \geq 2$, we check that χ is primitive if and only if it is injective on the subgroup $\mathbb{Z}/\langle 2^{k-2}\rangle$. For $k = 2$, the character is primitive if and only if it is not the principal character with modulus 4.

Gauss Sums for primitive Dirichlet characters

Fix a Dirichlet character χ of modulus m and we study $G(\chi, k)$ for various values of k. Recall that

$$G(\chi, k) = \sum_{a=1}^{m} \chi(a) \exp\left(2\pi\sqrt{-1}\frac{ak}{m}\right) = \sum_{\substack{a=1\\ \gcd(a,m)=1}}^{m} \chi(a) \exp\left(2\pi\sqrt{-1}\frac{ak}{m}\right)$$

since $\chi(a) = 0$ if $\gcd(a, m) \neq 1$.

We separate the calculation into two cases.

The case $\gcd(k, m) = 1$**.** Since ($k \bmod m$) is a unit in $\mathbb{Z}/\langle m\rangle$, the elements ($ak \bmod m$) are a permutation of the elements ($a \bmod m$).

Secondly, $\chi(a) = \overline{\chi}(k)\chi(ak)$ since $\chi(k)\overline{\chi}(k) = 1$. Thus, we get

$$\begin{aligned}
G(\chi, k) &= \sum_{a=1}^{m} \chi(a) \exp\left(2\pi\sqrt{-1}\frac{ak}{m}\right)\\
&= \sum_{a=1}^{m} \overline{\chi}(k)\chi(ak) \exp\left(2\pi\sqrt{-1}\frac{ak}{m}\right)\\
&= \overline{\chi}(k)\sum_{a=1}^{m} \chi(a) \exp\left(2\pi\sqrt{-1}\frac{a}{m}\right) = \overline{\chi}(k)G(\chi, 1)
\end{aligned}$$

For k co-prime to the modulus m we have $G(\chi, k) = \overline{\chi}(k)G(\chi, 1)$.

The case $\gcd(k, m) \neq 1$**.** Suppose that $q = \gcd(k, m) \neq 1$. We put $d = m/q$. Choose any b such that d divides $b - 1$. Then $m = qd$ divides $k(b - 1)$. Hence, $abk \bmod m = ak \bmod m$. This means

$$\exp\left(2\pi\sqrt{-1}\frac{abk}{m}\right) = \exp\left(2\pi\sqrt{-1}\frac{ak}{m}\right)$$

Now assuming $\gcd(b, m) = 1$, we can caculate the sum as follows.

$$\begin{aligned}
G(\chi, k) &= \sum_{a=1}^{m} \chi(a) \exp\left(2\pi\sqrt{-1}\frac{ak}{m}\right)\\
&= \sum_{a=1}^{m} \chi(ab) \exp\left(2\pi\sqrt{-1}\frac{abk}{m}\right) = \chi(b)\sum_{a=1}^{m} \chi(a) \exp\left(2\pi\sqrt{-1}\frac{ak}{m}\right)\\
&= \chi(b)G(\chi, k)
\end{aligned}$$

If $G(\chi,k)\neq 0$, this means that $\chi(b)=1$ for *any* integer b that satisfies $b \bmod d = 1$ and $\gcd(b,m)=1$.

It follows that $\chi_* : (\mathbb{Z}/\langle m\rangle)^* \to \mathbb{C}^*$ factors through $(\mathbb{Z}/\langle d\rangle)^*$.

In other words, if $G(\chi,k)\neq 0$ for some k such that $\gcd(k,m)\neq 1$, then χ is not primitive.

Turning this around, we see that if χ is a primitive Dirichlet character then $G(\chi,k)=0$ for all k such that $\gcd(k,m)\neq 1$. Note that the latter condition means that $\overline{\chi}(k)=0$.

Combining these two cases, we obtain the following.

> If χ is a primitive Dirichlet character, then $G(\chi,k)=\overline{\chi}(k)G(\chi,1)$ for all values of k.

Computation of $|G(\chi,1)|^2$

In this section χ is a primitive Dirichlet character with modulus m.

First of all, we make some observations about the sums of powers of a root of unity. For $k \bmod m$ non-zero,

$$\sum_{a=1}^{m}\exp\left(2\pi\sqrt{-1}\frac{ak}{m}\right)=\frac{\exp\left(2\pi\sqrt{-1}\frac{k}{m}\right)^m-1}{\exp\left(2\pi\sqrt{-1}\frac{k}{m}\right)-1}=0$$

We also note that

$$\sum_{a=1}^{m}\exp\left(2\pi\sqrt{-1}\frac{ak}{m}\right)=m \text{ for } k \bmod m=0$$

We use this in the following calculation.

$$\begin{aligned}
|G(\chi,1)|^2 &= G(\chi,1)\overline{G(\chi,1)}\\
&= G(\chi,1)\sum_{a=1}^{m}\overline{\chi}(a)\exp\left(2\pi\sqrt{-1}\frac{-a}{m}\right)=\sum_{a=1}^{m}G(\chi,a)\exp\left(2\pi\sqrt{-1}\frac{-a}{m}\right)\\
&= \sum_{a=1}^{m}\sum_{b=1}^{m}\chi(b)\exp\left(2\pi\sqrt{-1}\frac{ab}{m}\right)\exp\left(2\pi\sqrt{-1}\frac{-a}{m}\right)\\
&= \sum_{b=1}^{m}\chi(b)\sum_{a=1}^{m}\exp\left(2\pi\sqrt{-1}\frac{a(b-1)}{m}\right)=m\chi(1)=m
\end{aligned}$$

since the only term that survives in the second sum is when $b=1$ and in that case the value is m.

> If χ is a primitive Dirichlet character with modulus m, then $|G(\chi,1)|^2=m$.